

Family Name _____

First Name _____

Student Number

--	--	--	--	--	--	--	--	--	--

Venue _____

Seat Number _____



No electronic/communication devices are permitted.

No exam materials may be removed from the exam room.

Civil and Natural Resources Engineering EXAMINATION

Mid-Year Examinations, 2026

ENCI335-26S1 (C) Structural Analysis and Systems 1

For Examiner Use Only

Question Mark

Examination Duration: 150 minutes

Exam Conditions:

Closed book: Students may bring in writing implements only (and calculator if approved), no materials.

N/A - not restricted book exam.

Calculators with a 'UC' sticker permitted.

No exam materials may be removed from the exam room.

Materials Permitted in the Exam Venue:

None

Materials to be Supplied to Students:

1 x Gradescope Write-on exam paper

String to tie exam materials together

Instructions to Students:

- Total marks: ##
- Answer all questions. All numbers provided have consistent units. Formula sheets attached.
- Provide answers inside the provided boxed space only. If you find the boxed space insufficient, put a note, continue your answers on additional sheets, and attach them to the back of the exam book.

Total _____

Questions Start on Page 3

Question 1. (Total = ## marks)

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis.

[Answer]

[Answer]

Question 2. (Total = ## marks)

Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque.

[Answer]

[Answer]

Extra Answer Sheets

[Answer]

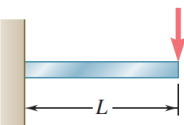
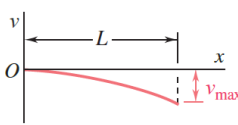
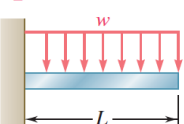
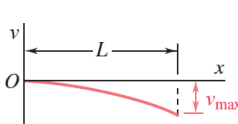
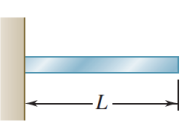
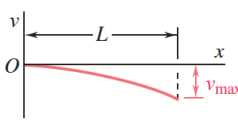
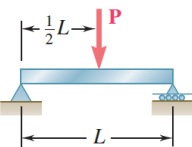
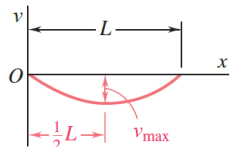
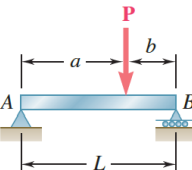
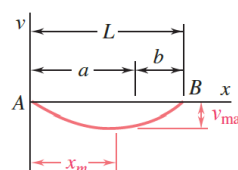
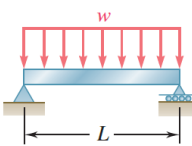
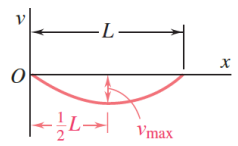
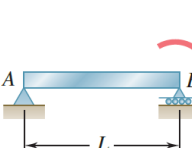
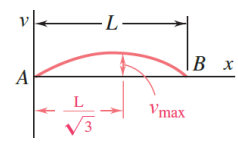
[Answer]

Formulas and Tables

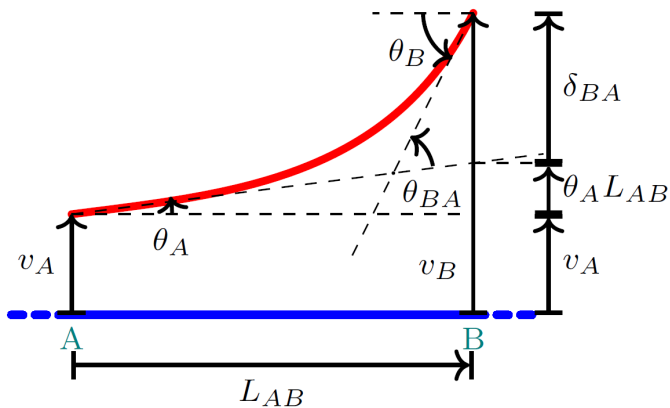
Degree of Static Indeterminacy

$$SI = n_R + n_{MF} - n_{JE} - n_H$$

Table of Beam Deflections

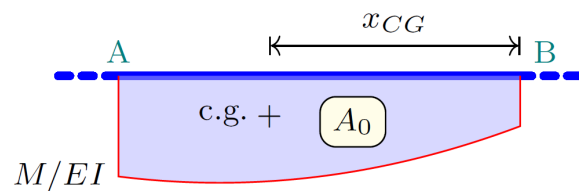
Beam and Loading	Elastic Curve	Maximum Deflection	Slope at End	Equation of Elastic Curve
1 		$-\frac{PL^3}{3EI}$	$-\frac{PL^2}{2EI}$	$v = \frac{P}{6EI}(x^3 - 3Lx^2)$
2 		$-\frac{wL^4}{8EI}$	$-\frac{wL^3}{6EI}$	$v = -\frac{w}{24EI}(x^4 - 4Lx^3 + 6L^2x^2)$
3 		$-\frac{M_B L^2}{2EI}$	$-\frac{M_B L}{EI}$	$v = -\frac{M_B}{2EI}x^2$
4 		$-\frac{PL^3}{48EI}$	$\pm \frac{PL^2}{16EI}$	For $x \leq \frac{1}{2}L$: $v = \frac{P}{48EI}(4x^3 - 3L^2x)$
5 		For $a > b$: $-\frac{Pb(L^2 - b^2)^{3/2}}{9\sqrt{3}EIL}$ at $x_m = \sqrt{\frac{L^2 - b^2}{3}}$	$\theta_A = -\frac{Pb(L^2 - b^2)}{6EIL}$ $\theta_B = +\frac{Pa(L^2 - a^2)}{6EIL}$	For $x < a$: $v = \frac{Pb}{6EIL}[x^3 - (L^2 - b^2)x]$ For $x = a$: $v = -\frac{Pa^2b^2}{3EIL}$
6 		$-\frac{5wL^4}{384EI}$	$\pm \frac{wL^3}{24EI}$	$v = -\frac{w}{24EI}(x^4 - 2Lx^3 + L^3x)$
7 		$\frac{M_B L^2}{9\sqrt{3}EI}$	$\theta_A = +\frac{M_B L}{6EI}$ $\theta_B = -\frac{M_B L}{3EI}$	$v = -\frac{M_B}{6EIL}(x^3 - L^2x)$

Moment-Area Theorems



$$v_B = v_A + \theta_A L_{AB} + \delta_{BA}$$

Relates the deflections at A & B



1st Moment-Area Theorem:

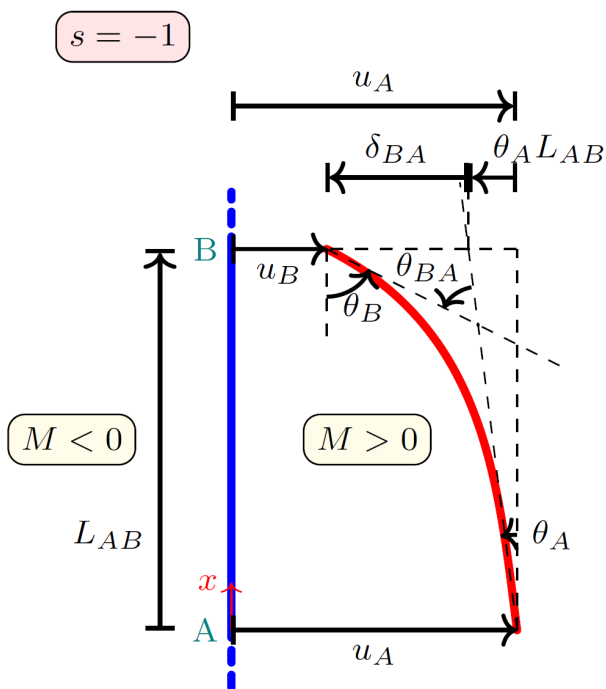
$$\theta_{BA} = \theta_B - \theta_A = \int_{L_{AB}} \frac{M}{EI} dx = A_0$$

2nd Moment-Area Theorem:

$$\delta_{BA} = \int_{L_{AB}} \frac{M}{EI} x' dx = A_0 \cdot x_{CG}$$

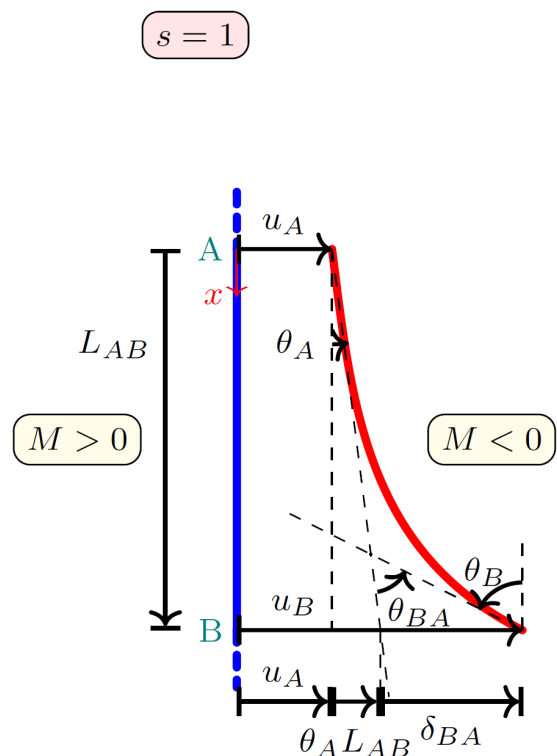
The change in θ from A to B:
 $\theta_{BA} =$ the M/EI area between A & B

The tangential deviation of B wrt. A:
 $\delta_{BA} =$ the first moment of the M/EI area about B



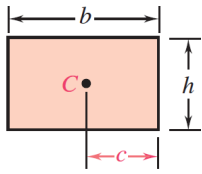
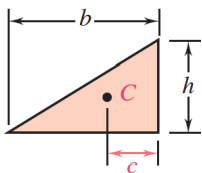
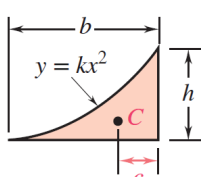
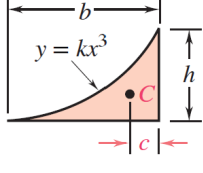
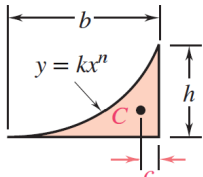
$$\theta_B = \theta_A + \theta_{BA}$$

$$u_B = u_A + (-1) \cdot (\theta_A L_{AB} + \delta_{BA})$$



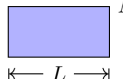
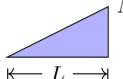
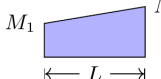
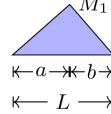
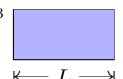
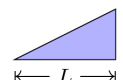
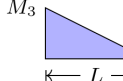
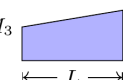
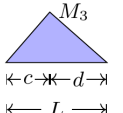
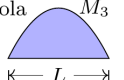
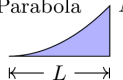
$$\theta_B = \theta_A + \theta_{BA}$$

$$u_B = u_A + 1 \cdot (\theta_A L_{AB} + \delta_{BA})$$

Shape		Area	c
Rectangle		bh	$\frac{b}{2}$
Triangle		$\frac{bh}{2}$	$\frac{b}{3}$
Parabolic spandrel		$\frac{bh}{3}$	$\frac{b}{4}$
Cubic spandrel		$\frac{bh}{4}$	$\frac{b}{5}$
General spandrel		$\frac{bh}{n+1}$	$\frac{b}{n+2}$

Energy

Table 1: Values of Product Integrals $\int_{x=0}^{x=L} M_Q M_P dx$ (adapted from Leet et al.)

$M_P \backslash M_Q$				
	$M_1 M_3 L$	$\frac{1}{2} M_1 M_3 L$	$\frac{1}{2} (M_1 + M_2) M_3 L$	$\frac{1}{2} M_1 M_3 L$
	$\frac{1}{2} M_1 M_3 L$	$\frac{1}{3} M_1 M_3 L$	$\frac{1}{6} (M_1 + 2M_2) M_3 L$	$\frac{1}{6} M_1 M_3 (L + a)$
	$\frac{1}{2} M_1 M_3 L$	$\frac{1}{6} M_1 M_3 L$	$\frac{1}{6} (2M_1 + M_2) M_3 L$	$\frac{1}{6} M_1 M_3 (L + b)$
	$\frac{1}{2} M_1 (M_3 + M_4) L$	$\frac{1}{6} M_1 (M_3 + 2M_4) L$	$\frac{1}{6} M_1 (2M_3 + M_4) L + \frac{1}{6} M_2 (M_3 + 2M_4) L$	$\frac{1}{6} M_1 M_3 (L + b) + \frac{1}{6} M_1 M_4 (L + a)$
	$\frac{1}{2} M_1 M_3 L$	$\frac{1}{6} M_1 M_3 (L + c)$	$\frac{1}{6} M_1 M_3 (L + d) + \frac{1}{6} M_2 M_3 (L + c)$	for $c \leq a$: $\left(\frac{1}{3} - \frac{(a-c)^2}{6ad} \right) M_1 M_3 L$
	$\frac{2}{3} M_1 M_3 L$	$\frac{1}{3} M_1 M_3 L$	$\frac{1}{3} (M_1 + M_2) M_3 L$	$\frac{1}{3} M_1 M_3 \left(L + \frac{ab}{L} \right)$
	$\frac{1}{3} M_1 M_3 L$	$\frac{1}{4} M_1 M_3 L$	$\frac{1}{12} (M_1 + 3M_2) M_3 L$	$\frac{1}{12} M_1 M_3 \left(3a + \frac{a^2}{L} \right)$

End of Examination